Thermodynamic description of the Ag–Sn system: a synthetic benchmark document

ORCHID synthetic benchmark (generated; no human authors)

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Abstract

This synthetic document describes the Ag–Sn binary system as calculated with pycalphad from the SGTE dataset AgSn-12Gie.tdb; every figure, table and stated value is generated, and the exact values are kept in a separate truth key. The highest invariant reaction, (Ag) + L → ζ, lies at 719 °C.

1. Introduction

The Ag–Sn system belongs to the 20 binaries of the ORCHID benchmark, chosen to cover diagram types such as peritectic. The thermodynamic description is the SGTE collection entry (Hallstedt, 2025), which reproduces the assessment of W. Gierlotka, J. Electron. Mater., 41, 86-108(2012). The calculations use pycalphad 0.11.2 at 1 atm.

Pure Ag melts at 962 °C. The melting point of Sn reproduced by the dataset is 232 °C.

2. Phase diagram

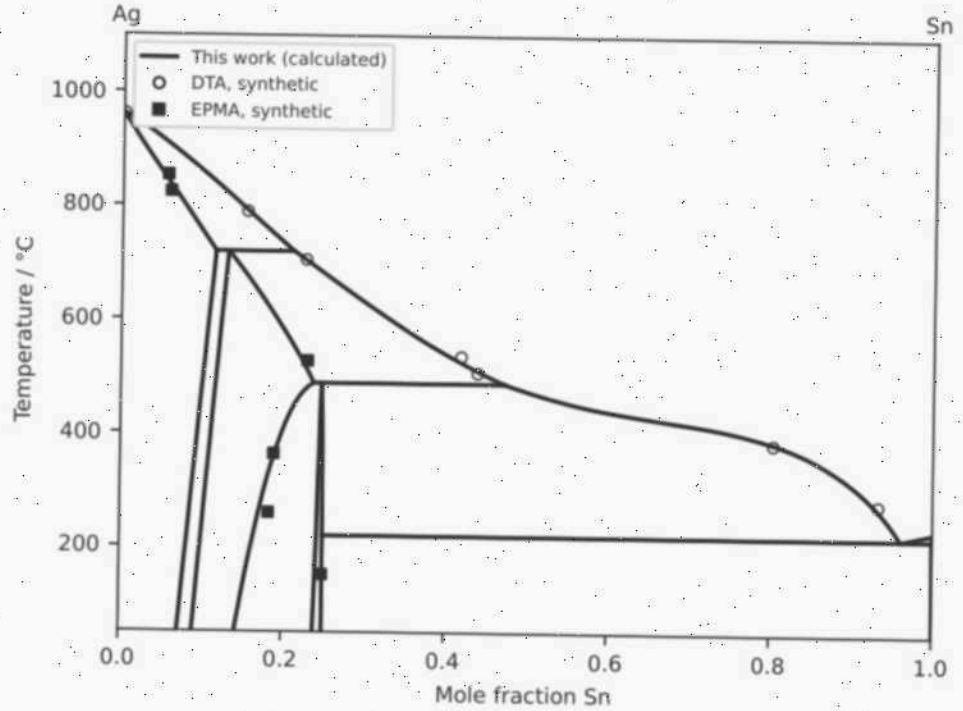


Fig. 1. Calculated Ag–Sn phase diagram with synthetic experimental-style points.

The dataset places a peritectic ((Ag) + L → ζ) at 719 °C; the composition of L is a mole fraction of 0.211. At 488 °C the peritectic ζ + L → Ag3Sn takes place; the L phase then contains x(Sn) = 0.473. The dataset places a eutectic (L → Ag3Sn + (Sn)) at 220 °C; the composition of L is a mole fraction of 0.961.

Table 1. Invariant reactions in the Ag–Sn system (calculated).

Reaction	Type	T / °C	Compositions (x(Sn))		
			phase 1	phase 2	phase 3
(Ag) + L → ζ	peritectic	719.3	0.115	0.211	0.131
ζ + L → Ag3Sn	peritectic	488.2	0.237	0.473	0.247
L → Ag3Sn + (Sn)	eutectic	219.8	0.961	0.250	0.999

3. Thermodynamic properties

At 1300 K the integral mixing enthalpy of the liquid has its minimum, −3147 J/mol, at x(Sn) = 0.220. The calculated enthalpy of formation of ζ at 298.15 K is −2.67 kJ/mol per mole of atoms.

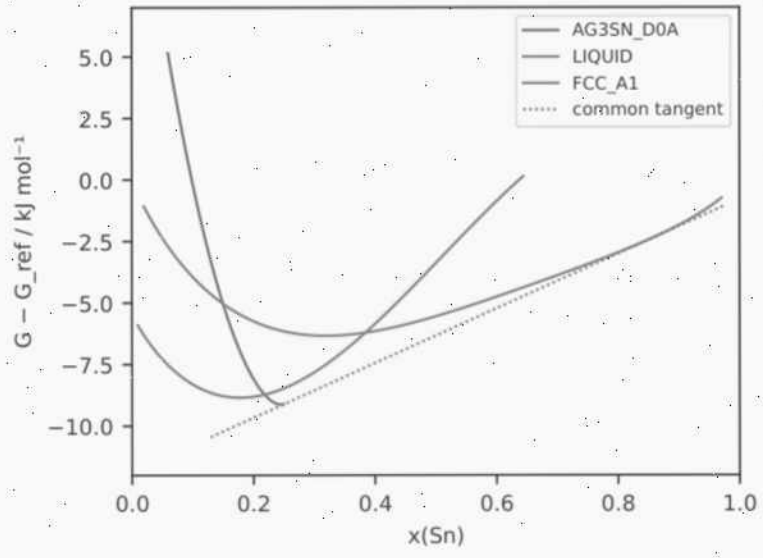


Fig. 2. Calculated Gibbs energy curves at 630 K.

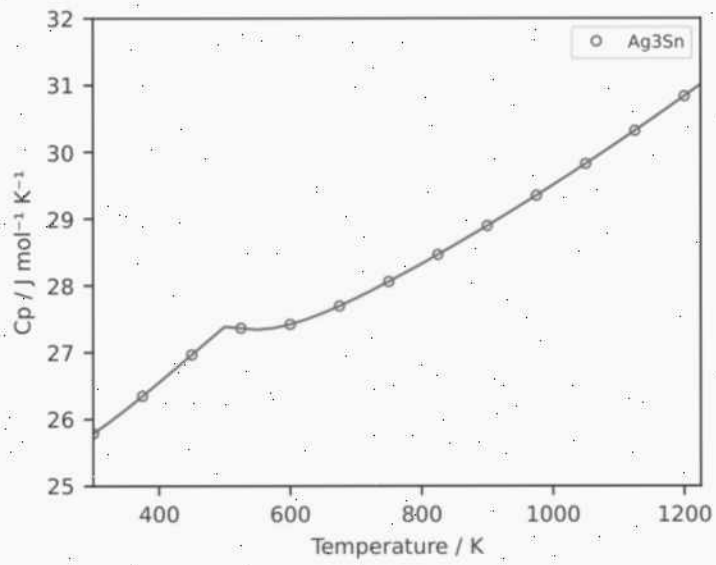


Fig. 3. Calculated heat capacity of Ag3Sn.

Table 2. Calculated formation properties of the intermediate phases (Ag–Sn).

		Calculated at 298.15 K, per mole of atoms	
Phase	**at.% Sn**	**ΔfH / kJ mol⁻¹**	**S / J mol⁻¹ K⁻¹**
Ag3Sn	24.4	-3.95	47.06
ζ	15.7	-2.67	47.12

4. Conclusions

This document is synthetic and exists only to test extraction tools; its truth key lists every boundary, field, invariant, plotted point, table cell and stated value.